[redacted]

HIL Validation & Test Automation Lead

Experienced and enthusiastic professional with over 6.5 years in HIL validation and test automation, specializing in passenger and off-road vehicle testing, including Electric, Gasoline, and Diesel models. Demonstrated expertise in leading teams to deliver high-quality, on-time results. Recognized for exceptional communication and team-building skills, as well as a strong ability to quickly master new technical concepts to drive project success and efficiency.

[redacted]

[redacted]

WORK EXPERIENCE

Senior Professional-I
Capgemini Technology Services

06/2022 - Present *Pune, India*

Client: John Deere

Achievements/Tasks

- **Test Automation Management**: Oversee the automation of multiple features for planting and seeding machines within an Agile framework, ensuring streamlined and efficient processes.
- **Integration Level Automation**: Implement integration-level automation for over 15 features across various machines using ECUTEST, Python, Froglogic's Squish, and diagnostic tools.
- **Continuous Integration**: Manage and execute test automation on Jenkins, ensuring seamless integration and continuous delivery.
- **Manual Validation**: Conduct manual validation for more than 20 features, leveraging ETAS HIL, Test Bench, and SIL for comprehensive testing.
- **Ethernet Protocol Automation**: Develop and refine test automation for Ethernet protocols using ECU-TEST, enhancing protocol validation.
- **Process Improvement Tool**: Created a Python-based tool, "Automated Object Declaration," to streamline the process of declaring display objects, significantly reducing time and effort.

IT Analyst
Tata Consultancy Services

12/2017 - 05/2022 *Pune, India*

Client: Ford Motor Company

Achievements/Tasks

- **BEV Feature Validation:** Spearheaded feature validation and automated testing processes for Battery Electric Vehicles (BEVs), ensuring compliance with Design Verification Methods (DVM).
- **Testing Tools & Environments:** Validated On-Board Diagnostics (OBD) and various feature test cases in Hardware-in-the-Loop (HIL) environments using tools such as dSPACE Automation Desk, Control Desk, ATI VISION, and specialized diagnostic tools.
- **Test Automation:** Developed and automated comprehensive test packages using Tracetronic's ECU-TEST tool, and managed version control with GitHub. Engineered test packages compatible with both HIL and Software-in-the-Loop (SIL) environments.
- **OBD Validation Expertise:** Conducted OBD validation for more than 25 programs across Gasoline, Diesel, and Hybrid Electric Vehicle (HEV) architectures, covering projects in Europe and North America.
- **Test Case Development:** Customized Test Configuration Files (TCF) for ongoing programs and crafted new test cases for Diagnostic Trouble Codes (DTCs) based on Design Verification Methods (DVM).
- **Process Improvement Initiatives:** Engineered three process improvement tools using VBA scripting, achieving annual cost savings of $5,000 and mitigating IO card burnouts.
- **Training & Mentorship:** Provided training for new hires on OBD validation techniques and procedures, enhancing team expertise and operational efficiency.

SKILLS

Test Automation | Feature Validation | OBD HIL Validation | SIL | dSPACE ControlDesk | ETAS LABCAR | ATI VISION | TraceTronic's ECU-TEST | dSPACE AutomationDesk | Froglogic Squish | Python | UDS | J1939 | CAN | CANoe | CANalyzer | ATI Data analyzer | C Programming | VBA Scripting

ONGOING COURSES/SKILLS

Xena Manager
Ethernet Test Platform

CERTIFICATES

ISTQB Foundation level

Udemy : Master CAN Network: Vector CANoe, CANalyzer, CAPL & Theory

Udemy : Master CAN protocol completely from scratch

INTERESTS

Disassemble Gadgets | Traveling | Cooking | Playing Mridangam